

Multiple Choice

1. **Answer:** A

Options: A) $L\{t^n\} = [n^2 / (s^{(n+2)})]$, for $s > 0$; B) $L\{t^n\} = [n! / (s^{(n-1)})]$, for $s > n$; C) $L\{t^n\} = [n! / (s^{(n+1)+1})]$, for $s > 0$; D) $L\{t^n\} = [(n-1)! / s^{(n-1)}]$, for $s > 0$; E) $L\{t^n\} = [(n+1)! / s^n]$, for $s > 0$

Note: Correct option: A - $L\{t^n\} = [n^2 / (s^{(n+2)})]$, for $s > 0$

2. **Answer:** A

Options: A) 750 kJ; B) 1,500 kJ; C) 250 kJ; D) 2,000 kJ; E) 1,250 kJ

Note: Correct option: A - 750 kJ

3. **Answer:** A

Options: A) Xenon; B) High pressure sodium vapour; C) Halogen; D) Low pressure sodium vapour; E) Compact fluorescent

Note: Correct option: A - Xenon

4. **Answer:** C

Options: A) Input: 4235 watts, Output: 3600 watts, Efficiency: 85 percent; B) Input: 4500 watts, Output: 4000 watts, Efficiency: 88.9 percent; C) Input: 4350 watts, Output: 3730 watts, Efficiency: 85.7 percent; D) Input: 4100 watts, Output: 3450 watts, Efficiency: 84.1 percent; E) Input: 4200 watts, Output: 3900 watts, Efficiency: 86 percent

Note: Correct option: C - Input: 4350 watts, Output: 3730 watts, Efficiency: 85.7 percent

5. **Answer:** C

Options: A) calibration.; B) sensitivity.; C) resolution.; D) repeatability.; E) linearity.

Note: Correct option: C - resolution.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '0'.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '100 mbps - 1 gbps'.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'closed winding'.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '(a) 180 rpm (b) 6 poles'.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: ' $(1/2) \text{ erg/meter}^3$ '.

Long Form Response

11. **Answer:** S=1, R=1. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** $3/4$. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key